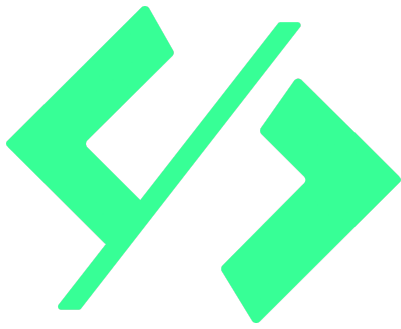




Linear Memory Reprogramming

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Difficulty: Medium-hard

Description

Your task is to overwrite a linear strip of memory using a special injection tool. The tool allows you to **overwrite any range** in memory with a **single uniform byte** (same character).

Each overwrite can target **any range**, but must contain **only one byte value**, and overwrites previous data.

Your objective:

Minimize the number of injections needed to exactly match the target memory string.

Examples

EX1

Input: `s = abcabc`

Output: `5`

Explanation:

- Print "aaaa" from index 0 to 3 this contiguous injection will count as 1 injection
- Print "b" at index 1 (overwrite a to get 'b') now u have "abaa" with injection number == 2
- Print "b" at index 4 now u have "abaab" with injection number == 3
- Print "c" at index 2 now u have "abcab" with injection number == 4
- Print "c" at index 5 now u have "abcabc" with injection number == 5

EX2

Input: s = abab

Output: 3

Explanation:

- Print "aaa" with injection number == 1
- Print at Index 1 "b" (overwrite a to get 'b') now u got "aba" with injection number == 2
- Print "b" at the index 3 now u got "abab" with injection number == 3

Constraints:

- $1 \leq s.length \leq 100$
- s consists of lowercase English letters.